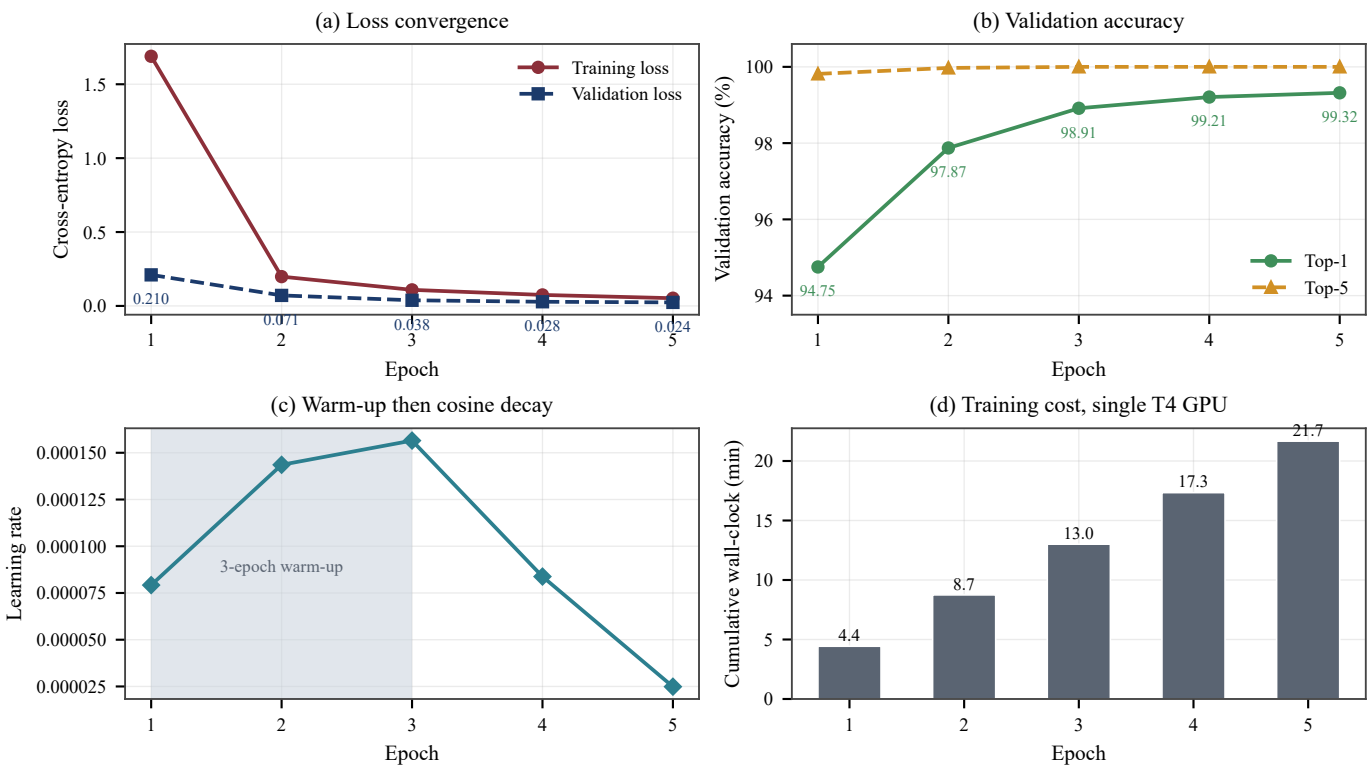
(a) Loss convergence
Training loss
Validation loss
Cross-entropy loss
1.5
1.0
0.5
0.0
0.210
0.071
0.038
0.028
0.024
1
2
3
4
5
Epoch
(b) Validation accuracy
Validation accuracy (%)
100
98
96
94
94.75
97.87
98.91
99.21
99.32
Top-1
Top-5
1
2
3
4
5
Epoch
(c) Warm-up then cosine decay
Learning rate
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
3-epoch warm-up
1
2
3
4
5
Epoch
(d) Training cost, single T4 GPU
Cumulative wall-clock (min)
21.7
17.3
13.0
8.7
4.4
20
15
10
5
0
1
2
3
4
5
Epoch